IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 23 February 2025, accepted 15 March 2025, date of publication 19 March 2025, date of current version 31 March 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3552842

RESEARCH ARTICLE

Bit-Mapping Based Security-Aware Energy-Efficient MAC Protocol for LoRaWAN

K. TUSHAR VIKASH, (Member, IEEE), GAURAV SUMAN, MANOJ TOLANI, (Senior Member, IEEE), AKSHAY TEJA TUMUNURI, (Member, IEEE), AND PANKAJ KUMAR, (Member, IEEE)

Department of Information and Communication Technology, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, Karnataka 576104, India

Corresponding authors: Manoj Tolani (manoj.tolani@manipal.edu) and Pankaj Kumar (pankaj.kumar@manipal.edu)

Open access funding provided by Manipal Academy of Higher Education, Manipal.

ABSTRACT Low energy consumption and high security are two key challenges in achieving efficient data transmission in LoRa-based wireless communication. The present research focuses on enhancing the efficiency and security of data transmission with low energy consumption in wireless sensor networks by introducing security-aware methods and features. A Bit Mapping-based channel allocation method is proposed, distinguishing between critical and non-critical nodes to ensure prioritized communication for time-sensitive data. A comprehensive mathematical model is developed to evaluate the system performance by enabling the analysis of energy consumption at the end nodes, gateways, and overall system levels. In addition, a security-aware feature is integrated into the application layer using Advanced Encryption Standard (AES)-128 and AES-256 encryptions, dynamically adjusting security levels based on the criticality of the sensor node while ensuring minimal impact on energy consumption. The channel access is optimized by prioritizing energy management according to the criticality of the information processed, ensuring energy-efficient and secure communication. The performance of the proposed protocol is analyzed with respect to standard TDMA-based LoRa Medium access Control (MAC) protocols. The result analysis proves the superiority of the proposed protocol.

INDEX TERMS LoRa, LoRaWAN, MAC, advanced encryption standard, energy consumption, TDMA, critical and normal nodes.

I. INTRODUCTION

There is a huge growth in demand for IoT, as it serves many purposes due to its inter-networking capabilities. IoT comes into play for various monitoring requirements and data transmission over long-range communications. The flexibility of Long Range Wide Area Networks (LoRaWAN) has led to its widespread adoption across various domains, such as smart cities, agriculture, healthcare, and industrial automation. In addition, its ability to provide long-range communication with minimal power consumption makes it particularly suitable for IoT applications that require extended operational lifetimes with limited energy resources. Many Low power Wide Area Networks(LPWAN) have evolved over time, such as LoRa [1], [2], SigFox [3], [4], Weightless [5], Random Phase Multiple Access (RPMA) [6], [7] and Narrow Band-IoT [8]. Among them, LoRaWAN, developed by the LoRa Alliance, has attracted many researchers in the past decade for its capacity for long-range communication with low power consumption. However, the increasing dependence on LoRaWAN for critical applications introduces significant challenges, particularly concerning data security and energy efficiency. With other attributes, security is an added advantage when using LoRaWAN for communication. The mathematical signal processing description for LoRa modulation and demodulation was presented in [9]. Various wireless communication options are examined in [11] concerning data security to assess their suitability based on the specific needs of the agricultural sector. Addressing the above challenges requires novel

The associate editor coordinating the review of this manuscript and approving it for publication was Yuan Gao.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

solutions to ensure secure, reliable, and energy-efficient communication, especially in scenarios involving sensitive or mission-critical data transmission.

In the present work, our main focus is to introduce a lightweight, security-aware, energy-efficient MAC protocol for long-range data transmission. Integrating robust encryption methods, such as Advanced Encryption Standard (AES), often increases computational overhead and sometimes creates a challenge in low-power IoT environments. In addition, the absence of an efficient security-aware MAC protocol for varying data priorities—such as those from critical and non-critical nodes—further complicates the issue. LoRaWAN faces inherent trade-offs between energy consumption and security protocols. Therefore, we have proposed a bit-mapping-based energy-efficient MAC protocol with application layer security to reduce energy consumption. The proposed method reduces energy consumption by efficiently allowing the slots based on the criticality of the data (sensor nodes). In addition, we have proposed and proved that additional application layer security features do not compromise energy consumption.

While security-based energy savings are present, their individual impact is significantly smaller compared to the energy savings achieved through bit-mapping-based slot allocation and optimal transmission scheduling. Thus, our analysis primarily focuses on the overall energy efficiency benefits of the protocol rather than isolating the minor impact of security-based optimizations. Additionally, the Contention Access Period (CAP) for bit-mapping adds a minor energy overhead, the results demonstrate that the energy savings from bit-mapping and adaptive slot assignment far outweigh this cost in most practical scenarios. This trade-off is also thoroughly analyzed in the result section.

During the Contention Access Period (CAP), normal nodes transmit control packets only within their pre-assigned subslots, which are allocated during the setup phase. While the CAP duration is short, nodes remain in active mode as the Tx/Rx switching time is higher than the CAP itself, making further energy savings impractical while ensuring minimal collision probability.

The synchronization overhead is also a concern in traditional TDMA method. Therefore, the proposed method utilizes an event-driven synchronization mechanism, where gateway-initiated beacon signals dynamically synchronize active nodes instead of maintaining continuous global synchronization. This adaptive approach significantly reduces synchronization energy costs and complexity, making it more suitable for large-scale and dynamic LoRa networks.

To address the above-discussed issue and to introduce an energy-efficient and security-aware MAC protocol, the existing methods are analyzed, and the research gap is identified for the proposed work. From literature study, it is evident that there is a need for a novel approach to address the trade-off between energy consumption and enhanced security in LoRa communication systems. The contribution and the novelty of the proposed method is discussed below:

- Bit Mapping-based (BM-based) channel allocation strategy is developed by distinguishing between critical and non-critical nodes during data transmission.
- A comprehensive mathematical model is developed to analyze energy consumption at end nodes, gateways, and the overall system level.
- A security-aware feature is introduced in the application layer based on the criticality of a node, ensuring that energy consumption is not compromised.
- Channel access is prioritized by optimizing energy management based on the criticality of the information processed by each node.

The rest of the paper is structured as described. Section II also contains the related studies and developments that have been done in this area of research so far. Section III deals with the Motivation and Novel ideas. Section IV discusses the simulation results obtained, and Section V discusses the conclusion and future scope of the study. We have used various abbreviations and acronyms. The TABLE 1 provides the list of acronyms used in the paper.

TABLE 1. List of acronyms.

Acronym	Description
AES	Advanced Encryption Standard
AES-CMAC	AES Cipher-Based Message Authentication Code
BM-LoRa	Bit-Map based LoRa Approach
CAM	Content Addressable Memory
CR	Code Rate
DMA	Direct Memory Access
FPGA	Field Programmable Gate Array
HMAC	Hash-based Message Authentication Code
LPADA	Low-Power AES Data Encryption Architecture
LPWAN	Low Power Wide Area Networks
LoRaWAN	Long Range Wide Area Networks
MAC	Message Authentication Code
RPMA	Random Phase Multiple Access
RSSI	Received Signal Strength Indicator
SHA	Secure Hash Algorithm
SF	Spreading Factor
TDMA	Time Division Multiple Access

II. RELATED STUDIES AND BACKGROUND

LoRaWAN communication has been extensively explored for its long-range and low-power capabilities, which have opened various research opportunities. A primary focus area is the security of these communications and ways to enhance them to maintain secure data exchange. References [14], [15], and [16] have presented various analysis and surveys for security issues and vulnerabilities on LPWAN networks such as LoRaWAN, Sigfox and NB-IoT.

In their study, Aras et al. [10] discuss security vulnerabilities in LoRa communication, highlighting significant security concerns in LoRaWAN and LoRa that can be easily exploited. Their findings emphasize the need for developers and

manufacturers to implement additional security measures to address these vulnerabilities. Eldefrawy et al. [13] conducted a security assessment of LoRaWAN v1.0 and v1.1 using the formal security analysis tool, Scyther. Their analysis revealed that LoRaWAN v1.0 exhibited several security vulnerabilities. However, these vulnerabilities were effectively mitigated in LoRaWAN v1.1, as confirmed by the Scyther model's evaluation. The security framework of LoRaWAN relies on AES-CMAC (AES Cipher-Based Message Authentication Code) and AES-CTR (AES Counter), both of which are implemented using the AES-128 algorithm. While data authentication and decryption at the gateway can be time-intensive, security remains a critical priority.

To address these security challenges, various researchers have put forward solutions through the study and research works. Dang et al. [12] developed an AES-128 core on an Field Programmable Gate Array (FPGA), integrating Direct Memory Access (DMA) to minimize CPU utilization during memory operations. This approach significantly reduced time consumption by offloading authentication to hardware, making the solution efficient and easily adaptable for other communication applications.

Zhang et al. [17] introduced a physical-layer encryption algorithm for LoRa, leveraging the Received Signal Strength Indicator (RSSI). This approach integrates a secret key extraction algorithm with a secret sharing method to enhance security. Secure Low Power Communication (SeLPC) method is proposed in [18], where power utilization during data encryption is reduced by minimizing the encryption cycles of AES scheme. Allamanda et al. [20] employed AES-256 encryption for secure data transmission in a LoRa-based forest fire monitoring system. Their research also utilized Secure Health Algorithm (SHA)-256 to ensure both data availability and integrity.

Iqbal and Iqbal [19] implemented AES encryption on the ESP32 platform to secure LoRa communication within micro-grids, incorporating a Message Authentication Code (MAC) for authentication purposes. McWeeney et al. [27] carried out a comprehensive analysis of payload confidentiality in point-to-point LoRa communication for smart grids. Their study leveraged the ESP32 integrated with AES cryptography to investigate the trade-offs and potential security vulnerabilities inherent in LoRa-based communication systems [30].

Tsai et al. [21] introduced a low-power AES architecture named Low-Power AES Data Encryption Architecture (LPADA), designed to reduce power consumption through the use of a low-power SBox, power gating techniques, and power management methods. The architecture integrates AES with CMAC and CTR, maintaining the same level of security as the original AES. However, LPADA significantly reduces power usage during encryption due to its optimized SBox design and power gating. The SBox incorporates Content Addressable Memory (CAM), enabling parallel searching and lowering power consumption during search operations. Additionally, power gating minimizes both dynamic and static power consumption by deactivating unused blocks.

Tsai et al. [22] introduced a method for secure and authenticated communication at the application layer between two servers by integrating the AES-128 encryption algorithm with an elliptic curve cryptographic approach. The proposed system generates session keys uniquely for each pair of communicating servers, thereby strengthening the confidentiality and integrity of the transmitted messages.

Thaenkaew et al. [23] conducted a comparative analysis of energy consumption, security improvements, and time delay when employing AES-192 and AES-256 encryption algorithms instead of AES-128 in the LoRaWAN protocol. The findings revealed that while larger encryption sizes increased costs and energy consumption due to the additional AES computation rounds, the impact was minimal compared to other significant energy consumers, such as time spent over the air.

Windya et al. [24] proposed a method to prevent sniffing in LoRa communication by combining AES-128 and AES-256 encryption with Hash-based Message Authentication Code (HMAC). Experimental results demonstrated that this approach effectively ensures communication integrity, confidentiality, and authenticity within the LoRa network.

Rakhaine et al. [25] provided a comprehensive analysis of LoRaWAN messaging devices, focusing on antenna design, a comparative evaluation of power consumption, and security protocols. Their findings indicate that the lowest power consumption was achieved by utilizing a combination of spreading factor 7, a bandwidth of 125 kHz, and a data rate of 21880 bps. Additionally, integrating the AES encryption protocol significantly improved security while optimizing the overall communication performance.

Abboud and Abdoun [26] introduce a novel cryptographic approach leveraging AES-256. Their comparative study highlights the benefits of AES-256 over AES-128-based protocols, particularly in terms of enhanced security, energy efficiency, and network throughput for participating devices. While the study underscores the improved security provided by AES-256 compared to AES-128, it also notes an increase in transmission time as a trade-off.

The literature study highlights the need for a novel approach to effectively balance the trade-off between energy consumption and enhanced security in LoRa communication systems. While existing works have explored various security mechanisms in LoRa networks, they primarily focus on enhancing encryption, authentication, and intrusion detection without considering their impact on energy consumption. The computational and communication overhead introduced by these security methods often leads to increased power consumption, which is a significant challenge in resource-constrained IoT environments.

To address this gap, our proposed BM-LoRa protocol integrates a security-aware, energy-efficient MAC design that

dynamically adjusts encryption levels based on node criticality while leveraging a bit-mapping-based slot allocation scheme to reduce unnecessary slot allocation and to reduce energy consumption. By doing so, BM-LoRa ensures that security enhancements do not compromise energy efficiency, making it more suitable for real-world LoRa deployments. To analyze this, we propose a Bit-Mapping-based security-aware energy-efficient LoRa (BM-LoRa) MAC protocol that optimizes slot allocation and security mechanisms. A mathematical model is developed in the next section to analyze the energy consumption dynamics and security impact, providing a comprehensive framework for evaluating the proposed approach.

III. MATHEMATICAL MODEL OF THE BM-LORA

The mathematical model of the proposed method is developed for energy-efficient secure data transmission. The proposed MAC frame structure and proposed protocol are discussed in the following subsections:

A. POWER RATINGS OF LORA NODES

The power ratings for the gateway and end nodes are based on the transceivers SX1301 (gateway) [31] and SX1276 (end nodes) [32]. The power consumption for different states is defined as follows:

- P_{tx}: Transmission power
- P_{rx}: Reception power
- P_{s}: Sleep power
- P_{sb}: Standby power

B. PACKET STRUCTURE AND LORA PARAMETERS

The packet length is influenced by the preamble, PHY header, payload size, and CRC. The total packet length T_{len} is calculated as:

$$T_{\text{len}} = \text{Preamble} + \text{Mandatory Preamble} + \text{PHY Header} + \text{Payload} + \text{CRC} \tag{1}$$

The control packet length (C_{Tlen}) is:

$$C_{T_{\text{len}}} = \text{Preamble Length} + \text{Payload Length} \tag{2}$$

A packet consists of preamble symbols, Mandatory Preamble (sync words and SFD symbols), and a variable number of data chirps, with parameters such as Code Rate (CR), Implicit Header (IH), low data rate optimization (DE), and CRC.

The PHY header can be omitted with IH. The packet size is limited to 255 bytes.

C. LORA DATA RATE CALCULATION

The data rate for LoRa is determined by the Spreading Factor (SF), Bandwidth (BW), and Coding Rate (CR). The symbol duration T_{symbol} and data rate R_{data} are calculated as follows:

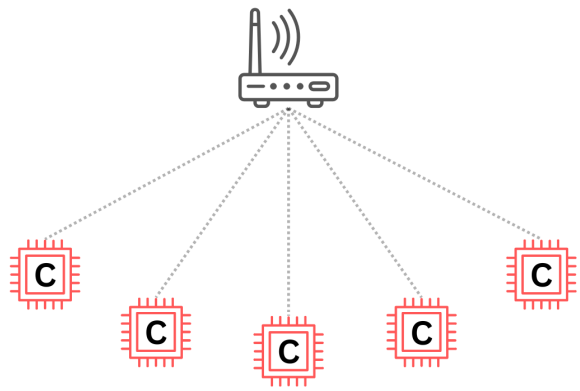

FIGURE 1. Scenario 1 - All nodes are designated as critical.

$$T_{\text{symbol}} = \frac{2^{\text{SF}}}{\text{BW}} \tag{3}$$

$$R_{\text{data}} = \frac{\text{SF} \times \text{CR}}{T_{\text{symbol}}} = SF \times CR \times \frac{BW}{2^{SF}} \tag{4}$$

The SF determines the symbol duration, where a higher SF decreases the data rate but enhances the range and signal robustness. Conversely, a lower SF increases the data rate but reduces the range.

D. LORAWAN ARCHITECTURE

1) SCENARIO 1: ALL CRITICAL NODES

In this case, we consider all nodes as critical, meaning they are responsible for transmitting essential data.

While traditional LoRaWAN networks utilize the CSMA/CA protocol at the MAC layer, we propose the use of Time Division Multiple Access (TDMA) to allocate dedicated time slots to each critical node. This ensures reliable, collision-free communication within the network.

2) SCENARIO 2: MIXED CRITICAL AND NORMAL NODES

In Scenario 2, the network consists of both critical (N_c) and normal nodes (N_n). The critical (N_c) and normal nodes (N_n) are determined based on the parameter c. A critical node in LoRa-based networks refers to a sensor node responsible for transmitting time-sensitive, high-priority, or security-sensitive data. Examples include industrial fault detection sensors, healthcare monitoring devices, environmental hazard sensors, and security surveillance nodes, which require prioritized transmission and enhanced security measures. Additionally, the normal nodes are further classified according to the probability p of having data to transmit:

$$N_c = N \times c, \quad N_n = N \times (1 - c) \tag{5}$$

$$N_{n1} = p \times N_n, \quad N_{n2} = (1 - p) \times N_n \tag{6}$$

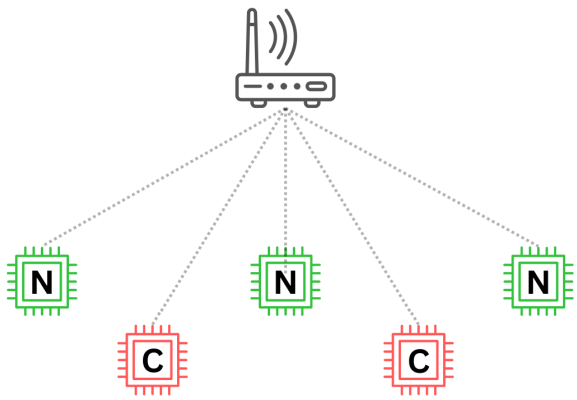

FIGURE 2. Scenario 2 - A subset of nodes are critical, while the rest are normal.

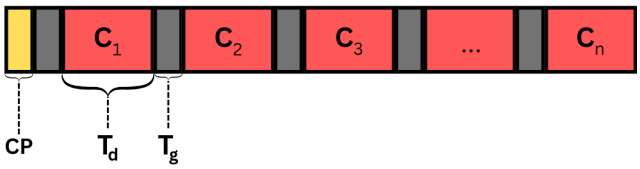

FIGURE 3. Scenario 1 - MAC Frame Structure.

E. MAC FRAME STRUCTURE

1) SCENARIO 1: ALL CRITICAL NODES

Here, *CP* represents the contention period, during which the gateway sends a control packet to initialize the frame and allocate slots for the nodes. T_{d} denotes the duration of the data slot, which is determined by the payload and the data rate (considering the SF and CR) and T_{g} refers to the guard time, which is the time interval between two data slots.

2) SCENARIO 2: MIXED CRITICAL AND NORMAL NODES

In this context, N_c denotes the number of critical nodes, while N_n represents the total number of normal nodes. N_{n1} refers to the normal nodes that have data to transmit.

The gateway sends a Control Packet (*CP*) to initialize the frame and allocate slots for the critical nodes. Each critical node is then assigned a dedicated TDMA slot for data transmission. Post the transmission of the critical nodes, the normal nodes utilize the Contention Access Period (*CAP*) to send control packets and request data slots, which are allocated based on the received requests during the *CAP*.

F. SECURITY ANALYSIS

1) SCENARIO 1: ALL CRITICAL NODES

In this scenario, we implement AES-256 encryption for all critical nodes, ensuring stronger security than AES-128. The results from the study by Munoz et al. [28] show that while AES-256 requires more energy, the difference in energy consumption between AES key sizes is relatively small. For example, increasing from AES-128 to AES-256 results in a 22.8% increase in energy consumption for software implementations (on CYW board [29]), but this increase is minimal, measured in the range of a few microjoules. The impact on overall system energy consumption is, therefore, negligible, especially when considering the robust security benefits of AES-256. The selective use of AES-128 for non-critical nodes reduces computational overhead and encryption-induced latency, ensuring that energy efficiency and real-time performance are maintained without compromising security for critical data transmissions.

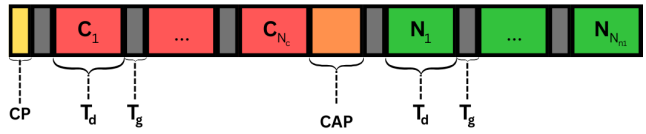

FIGURE 4. Scenario 2 - MAC Frame Structure.

2) SCENARIO 2: MIXED CRITICAL AND NORMAL NODES

For this mixed-node scenario, critical nodes use AES-256 encryption with TDMA for collision-free communication, while normal nodes use AES-128 encryption. According to Munoz et al. [28], the energy difference between AES-128 and AES-256 is small and within the range of a few microjoules, especially when buffer sizes increase. The study shows that while AES-256 consumes slightly more energy than AES-128, the total energy overhead is minimal, making it suitable for critical nodes without significantly affecting the overall system's energy efficiency.

G. ENERGY EFFICIENCY ANALYSIS

1) SCENARIO 1: ALL CRITICAL NODES

During the Control Packet (CP) phase, the gateway broadcasts a beacon message to announce the slot allocations for the frame. For instance, in the first time slot with a duration of T_d, node C_1 will transmit its data, while the remaining nodes will stay in sleep mode. This will occur for N such slots. During the guard time T_g, all nodes will enter standby mode. The gateway and end nodes will alternate between active and inactive states based on the number of cycles and the duty cycle. The energy consumption during the active period for both the gateway and end nodes is computed based on the following Eq 7 and 8:

The first term in Eq 7 represents the energy consumed by the gateway to transmit the CP, while the second term accounts for the reception energy of the gateway as it receives data from all N nodes over their allocated slots. The first term in Eq 8 represents the energy required for each node to receive the CP. The second term accounts for the energy required for a node to transmit data during its assigned slot. The third term considers the energy consumed by inactive nodes in sleep mode while waiting for their slots, while the final term represents the energy consumed during the guard time. During the Sleep Period, the Energy Consumption is as follows in Eq 9 and 10. Each individual terms of the energy consumption of all the equations are shown in Table 2.

TABLE 2. Definition of individual multiplication terms in equations.

Symbol	Definition
T_{cp}	Duration of the Control Packet (CP) transmission
T_{d}	Duration of data transmission slot for each node
T_{g}	Guard time between two consecutive slots
T_{cap}	Duration of the Contention Access Period (CAP)
P_{txg}	Transmission power of the gateway during CP transmission
P_{rxg}	Reception power of the gateway while receiving data
P_{rxn}	Reception power of the end nodes during CP reception
P_{txn}	Transmission power of end nodes during data transmission
P_{sn}	Power consumption of end nodes in sleep mode
P_{sbn}	Standby power consumption of end nodes during the guard time
P_{sg}	Sleep mode power consumption of the gateway
Multiplication Term	**Definition**
$T_{\text{cp}} \times P_{\text{txg}}$	Energy consumed by the gateway to transmit the Control Packet (CP)
$N \times (T_{\text{d}} + T_{\text{g}}) \times P_{\text{rxg}}$	Energy consumed by the gateway to receive data from all nodes
$N \times T_{\text{cp}} \times P_{\text{rxn}}$	Energy consumed by end nodes to receive the CP from the gateway
$N \times T_{\text{d}} \times P_{\text{txn}}$	Energy consumed by end nodes to transmit data in their allocated slots
$(N-1) \times T_{\text{d}} \times P_{\text{sn}}$	Energy consumed by inactive nodes in sleep mode while waiting for their slots
$N \times T_{\text{g}} \times P_{\text{sbn}}$	Energy consumed by end nodes in standby mode during the guard time
$N_c \times P_{\text{rxg}} \times (T_{\text{d}} + T_{\text{g}})$	Energy consumed by the gateway receiving data from critical nodes
$N_n \times P_{\text{rxg}} \times (T_{\text{d}} + T_{\text{g}})$	Energy consumed by the gateway receiving data from normal nodes
$T_{\text{cap}} \times P_{\text{rxg}}$	Energy consumed by the gateway during the Contention Access Period (CAP)
$N_c \times T_{\text{cp}} \times P_{\text{rxn}}$	Energy consumed by critical nodes while receiving the CP
$1 \times T_{\text{d}} \times P_{\text{txn}}$	Energy used by a single node to transmit data in its slot
$(N_c - 1) \times T_{\text{d}} \times P_{\text{sn}}$	Energy consumed by idle critical nodes in sleep mode
$N_n \times (T_{\text{d}} + T_{\text{g}}) \times P_{\text{sn}}$	Energy consumed by normal nodes in sleep mode
$N_{n1} \times P_{\text{txn}} \times T_{\text{cap}}$	Energy consumed by normal nodes transmitting in CAP
$N_{n2} \times P_{\text{sn}} \times T_{\text{cap}}$	Energy consumed by non-transmitting normal nodes in CAP
$N_c \times P_{\text{sn}} \times T_{\text{cap}}$	Energy consumed by critical nodes in CAP
$N_{n1} \times (1 \times P_{\text{txn}} \times T_d)$	Energy consumed by transmitting normal nodes
$(N_{n1} - 1) \times P_{\text{sn}} \times T_d$	Energy consumed by idle normal nodes in sleep mode
$N_{n1} \times P_{\text{sn}} \times T_g$	Energy consumed by normal nodes during guard time
$N_{n2} \times P_{\text{sn}} \times (T_d + T_g)$	Energy consumed by non-transmitting normal nodes in sleep mode
$N_c \times (T_d + T_g) \times P_{\text{sn}}$	Energy consumed by critical nodes in sleep mode during idle periods

2) SCENARIO 2: MIXED CRITICAL AND NORMAL NODES

During the Control Packet *CP* phase, the gateway broadcasts a message to the critical nodes, informing them of their allocated transmission slots. The critical nodes then transmit their data in accordance with the assigned slots while the normal nodes remain in sleep mode. In the Contention Access Period *CAP*, normal nodes with data to transmit send control packets to request a transmission slot. For example, in the first slot, node N_1 transmits its data, while the other nodes remain in sleep mode. During the guard time T_g, all normal nodes that did not respond in the CAP transition to standby mode. Throughout the process, critical nodes and any normal nodes that did not request a slot during the CAP remain in sleep mode. The energy consumption equations for the gateway and end nodes are given in Eq 11 and 12:

$$E_{TDMA_G} = (T_{\text{cp}} \times P_{\text{txg}}) + N \times ((T_{\text{d}} + T_{\text{g}}) \times P_{\text{rxg}}) \tag{7}$$

$$E_{TDMA_{EN}} = N \times \left[(T_{\text{cp}} \times P_{\text{rxn}}) + (T_{\text{d}} \times P_{\text{txn}}) + (N-1) \times (T_{\text{d}} \times P_{\text{sn}}) + (N \times T_{\text{g}} \times P_{\text{sbn}})\right] \tag{8}$$

$$E_{TDMA_G} = (T_{\text{cp}} \times P_{\text{sg}}) + N \times ((T_{\text{d}} + T_{\text{g}}) \times P_{\text{sg}}) \tag{9}$$

$$E_{TDMA_{EN}} = N \times \left[(T_{\text{cp}} \times P_{\text{sn}}) + N \times ((T_{\text{d}} + T_{\text{g}}) \times P_{\text{sn}})\right] \tag{10}$$

$$E_{BM_G} = (T_{\text{cp}} \times P_{\text{txg}}) + N_c \times \left(P_{\text{rxg}} \times (T_{\text{d}} + T_{\text{g}})\right) + N_n \times \left(P_{\text{rxg}} \times (T_{\text{d}} + T_{\text{g}})\right) + (T_{\text{cap}} \times P_{\text{rxg}}) \tag{11}$$

$$\begin{aligned} E_{BM_{EN}} = {} & (N_c \times T_{\text{cp}} \times P_{\text{rxn}}) + N_c \times [(1 \times T_{\text{d}} \times P_{\text{txn}}) \\ & + ((N_c - 1) \times T_{\text{d}} \times P_{\text{sn}}) + (N_n \times (T_{\text{d}} + T_{\text{g}}) \times P_{\text{sn}})] \\ & + \left(N_{n1} \times P_{\text{txn}} \times T_{\text{cap}}\right) + \left(N_{n2} \times P_{\text{sn}} \times T_{\text{cap}}\right) + (N_c \\ & \times P_{\text{sn}} \times T_{\text{cap}}) + N_{n1} \times [(1 \times P_{\text{txn}} \times T_d) \\ & + ((N_{n1} - 1) \times P_{\text{sn}} \times T_d) + \left(N_{n1} \times P_{\text{sn}} \times T_g\right) \\ & + \left(N_{n2} \times P_{\text{sn}} \times (T_d + T_g)\right) + \left(N_c \times (T_d + T_g) \\ & \times P_{\text{sn}})] \end{aligned} \tag{12}$$

$$E_{BM_G} = (T_{\text{cp}} \times P_{\text{sg}}) + N_c \times (P_{\text{sg}} \times (T_{\text{d}} + T_{\text{g}})) + N_n \times (P_{\text{sg}} \times (T_{\text{d}} + T_{\text{g}})) + (T_{\text{cap}} \times P_{\text{sg}}) \tag{13}$$

$$\begin{aligned} E_{BM_{EN}} = {} & (N_c \times T_{\text{cp}} \times P_{\text{sn}}) + N_c \times [N \times P_{\text{sn}} \times (T_{\text{d}} + T_{\text{d}})] \\ & + (N_{n1} \times P_{\text{sn}} \times T_{\text{cap}}) + (N_{n2} \times P_{\text{sn}} \times T_{\text{cap}}) \\ & + (N_c \times P_{\text{sn}} \times T_{\text{cap}}) + N_{n1} \\ & \times [(N \times P_{\text{sn}} \times (T_{\text{d}} + T_{\text{d}}))] \end{aligned} \tag{14}$$

These equations include the power consumption during the *CP*, *CAP* phase, and data transmission, considering both critical and normal nodes. During the Sleep Period, the Energy Consumption is as given in Eq 13 and 14:

The duty cycle influences the energy consumption of nodes during active and sleep periods. The total energy during the active period $E_{\text{Total Active}}$ and sleep period $E_{\text{Total Sleep}}$ during Number of cycles (N_{cycles}) are:

$$E_{\text{Total Active}} = N_{\text{cycles}} \times \text{Duty Cycle} \times E_{\text{Active}} \tag{15}$$

$$E_{\text{Total Sleep}} = N_{\text{cycles}} \times (1 - \text{Duty Cycle}) \times E_{\text{Sleep}} \tag{16}$$

The total energy consumption is the sum of these two components, evaluated over multiple duty cycles.

$$E_{\text{Total}} = E_{\text{Total Sleep}} + E_{\text{Total Active}} \tag{17}$$

IV. SIMULATION RESULTS AND DISCUSSION

The performance analysis of the proposed method is conducted using MATLAB R2024b to assess and compare the energy consumption at the end node, the gateway, and the overall system energy usage of the proposed Bit-Mapped LoRa MAC protocol in comparison to the TDMA-based strategy, including the evaluation of security mechanisms such as AES-128 and AES-256 encryption, ensuring that the proposed method remains energy-efficient while maintaining high security for critical nodes. The power ratings of the SX1301 (gateway) [31] and SX1276 (end nodes) [32] are used to develop a simulation scenario to prove the superiority of the proposed method [30]. These transceivers are widely used in LoRa deployments, ensuring that the power consumption calculations align with real-world implementations and reflect the typical operational conditions in IoT-based LoRa networks.

The analysis is organized into the following subsections to clearly highlight the advantages of the Bit-Mapped approach over the TDMA-based strategy for energy consumption.

To ensure compliance with EU standards and optimize performance, BM-LoRa operates with a spreading factor (SF) of 12, providing long-range and robust signal transmission. The bandwidth (BW) is set to 125 kHz, which is standard for LoRaWAN deployments, balancing data rate and coverage. The coding rate (CR) is fixed at 4/8, ensuring effective error correction while maintaining spectral efficiency. However, for comparative performance analysis in Case-3, the SF is varied to analyze its impact on energy consumption and network efficiency.

The parameters used for simulation analysis are given in Table 3. These parameters play a crucial role in determining the energy efficiency of the proposed BM-LoRa protocol and provide a structured approach to analyzing the trade-offs between security, energy consumption, and transmission reliability.

A. CASE 1: PROBABILITY VARIATION

In this scenario, the probability of data availability at the node is varied to evaluate energy consumption.

FIGURE 5 illustrates the Gateway's energy consumption. As anticipated, the BM-LoRa-based approach consumes less energy than the TDMA approach. Additionally, it is observed that the gateway's energy consumption remains constant in the BM-LoRa approach, regardless of the probability of data generation.

FIGURE 6 shows the total energy consumption of the end nodes. It shows that as the probability of data availability at the end nodes increases, overall energy consumption rises, particularly with a growing number of end nodes. Again, the BM-LoRa approach consistently exhibits significantly lower energy consumption compared to the TDMA-based strategy. However, under conditions of high data availability and a large number of end nodes, the TDMA-based strategy proves to be more energy-efficient than BM-LoRa. BM-LoRa achieves energy efficiency through dynamic slot allocation, activating only source nodes and keeping non-source nodes in sleep mode. This is highly effective under low-to-moderate data availability but becomes less efficient in high-traffic scenarios due to CAP overhead. Also, TDMA, with its fixed slot allocation, is inefficient for low traffic but outperforms BM-LoRa when all nodes are actively transmitting. BM-LoRa's security-aware slot allocation further slightly optimizes energy use, but its advantage reduces in dense traffic.

The overall energy consumption is analyzed in **FIGURE 7**, which illustrates that increasing the probability of data availability at the end node results in higher overall energy consumption, especially as the number of end nodes rises. Notably, the BM-LoRa approach demonstrates significantly lower energy consumption than the TDMA-based strategy. However, under conditions of high data availability and a large number of end nodes, the TDMA-based strategy consumes less energy than the BM-LoRa approach.

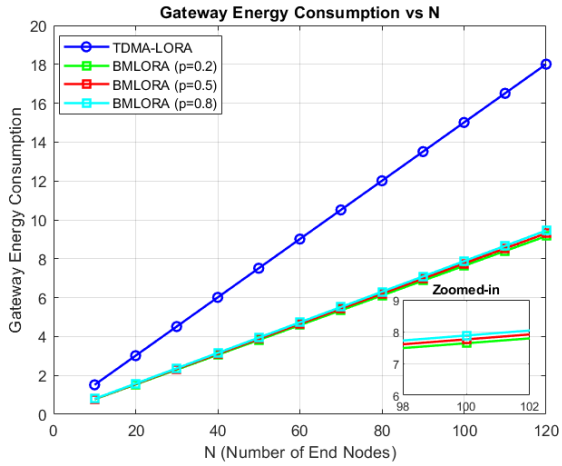

FIGURE 5. Gateway energy consumption(μJ) vs N (Variation in Probability).

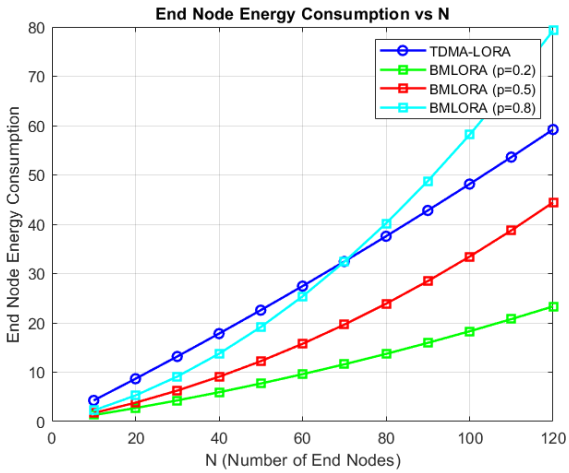

FIGURE 6. End node energy consumption(μJ) vs N (Variation in Probability).

TABLE 3. Simulation parameters for performance analysis.

Category	Parameter	Formula	Units
Gateway Power Ratings	P_{txg}	$(7.5 \times 10^{-3})A \times 3.3V$	W
	P_{rxg}	$(7.5 \times 10^{-3})A \times 3.3V$	W
	P_{sg}	$(1 \times 10^{-6})A \times 3.3V$	W
End Node Power Ratings	P_{txn}	$(20 \times 10^{-3})A \times 3.3V$	W
	P_{rxn}	$(11.5 \times 10^{-3})A \times 3.3V$	W
	P_{sbn}	$(1.8 \times 10^{-3})A \times 3.3V$	W
	P_{sn}	$(0.6 \times 10^{-6})A \times 3.3V$	W
Packet Values	Preamble	$6 \times SF$	–
	Mandatory_Preamble	$4.25 \times SF$	–
	PHY_Header	1.5×8	bits
	CRC	2×8	bits
LoRa Parameters	BW	(125×10^{-3})	MHz
	CR	$4/8$	–
	Symbol_Duration	$2^{SF}/BW$	seconds
Traffic Generation Parameters	p	Event probability parameter	0.5 (default)
	c	Critical node probability	0.5 (default)

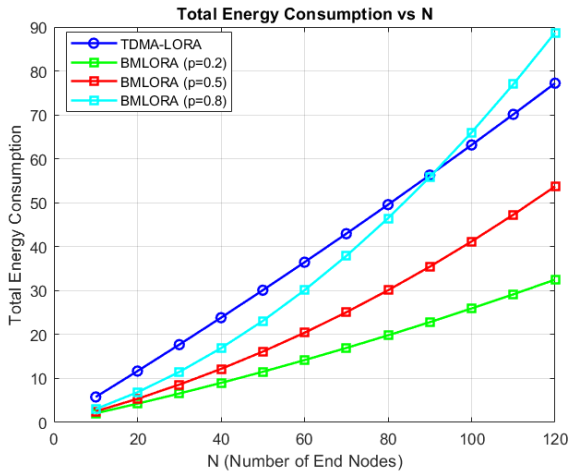

FIGURE 7. Total energy consumption(μJ) vs N (Variation in Probability).

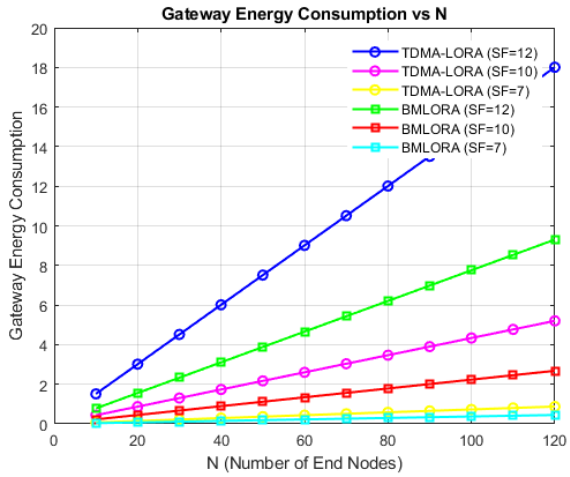

FIGURE 8. Gateway energy consumption(μJ) vs N (Variation in SF).

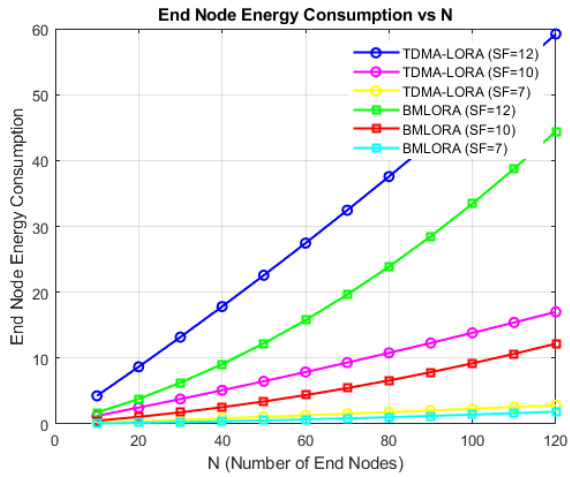

FIGURE 9. End node energy consumption(μJ) vs N (Variation in SF).

In the context of the BM-LoRa and TDMA-based strategies, we observe that AES-256 encryption increases energy consumption slightly compared to AES-128, especially as the probability of data availability rises. However, the increase in energy consumption due to AES-256 remains negligible when compared to the total energy consumed by the system. The BM-LoRa approach continues to be more energy-efficient than the TDMA-based strategy, and the energy difference between AES-128 and AES-256 does not significantly affect the overall energy savings. Thus, while AES-256 offers enhanced security, its energy cost is still marginal compared to the benefits of the BM-LoRa protocol.

B. CASE 2: VARIATION OF THE SPREADING FACTOR

In this scenario, the Spreading Factor is varied to analyze the energy consumption with respect to the number of end nodes.

FIGURE 8 presents the gateway's energy consumption, revealing a similar pattern where BM-LoRa consistently uses less energy than the TDMA-based approach.

Meanwhile, **FIGURE 9** illustrates the total energy consumption of the end nodes for both BM-LoRa and TDMA, across different spreading factors. It is evident that BM-LoRa is more energy-efficient in all scenarios.

FIGURE 10 highlights that the total energy consumption with the BM-LoRa approach is consistently lower than with

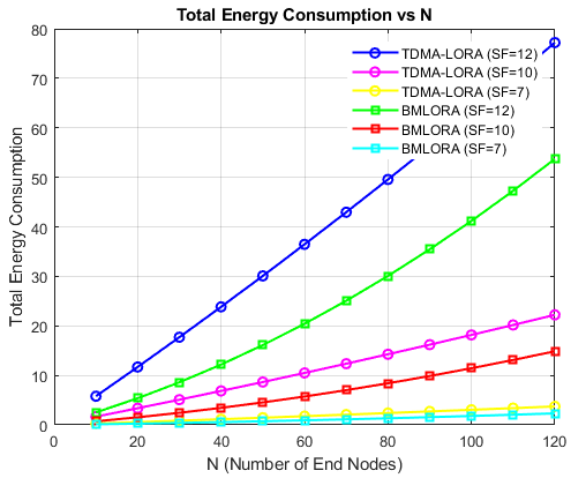

FIGURE 10. Total energy consumption(μJ) vs N (Variation in SF).

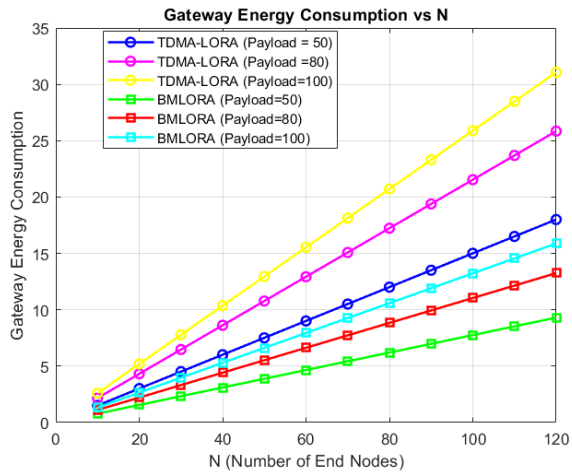

FIGURE 11. Gateway energy consumption(μJ) vs N (Variation in Payload).

the TDMA-based strategy for any given spreading factor. The difference in the slopes of the plots reflects the energy savings achieved by using BM-LoRa over TDMA. Notably, these savings become more pronounced as the spreading factor increases.

When varying the spreading factor, the energy consumption of both AES-128 and AES-256 increases, but the BM-LoRa approach remains more energy-efficient than TDMA-based solutions, even with the higher energy overhead of AES-256. The energy consumption differences between AES-128 and AES-256 are minimal and do not alter the overall trend. As the spreading factor increases, the added overhead of AES-256 encryption does result in a slight increase in energy usage, but the total energy consumption is still dominated by the lower energy usage of the BM-LoRa approach compared to TDMA.

C. CASE 3: VARIATION IN PAYLOAD LENGTH

In this case, the Payload length is varied to analyze the energy consumption with respect to the number of end nodes.

FIGURE 11 depicts the gateway's energy consumption, showing a consistent trend where BM-LoRa consumes less energy than the TDMA-based approach.

Similarly, **FIGURE 12** highlights the total energy consumption of end nodes for both BM-LoRa and TDMA across varying payload lengths, clearly demonstrating that BM-LoRa is more energy-efficient in all cases. Similarly, as the number of end nodes rises, the BM-LoRa-based approach (Payload = 100) shows energy consumption similar to that of the TDMA-based approach (Payload = 50), particularly when there is a 100% increase in throughput.

FIGURE 13 illustrates that the total energy consumption with BM-LoRa for a given payload size is consistently lower than that of the TDMA-based approach. The difference in the plot slopes represents the energy savings achieved with BM-LoRa compared to TDMA, and these savings become increasingly significant as the payload size grows. Furthermore, as the number of end nodes increases, the

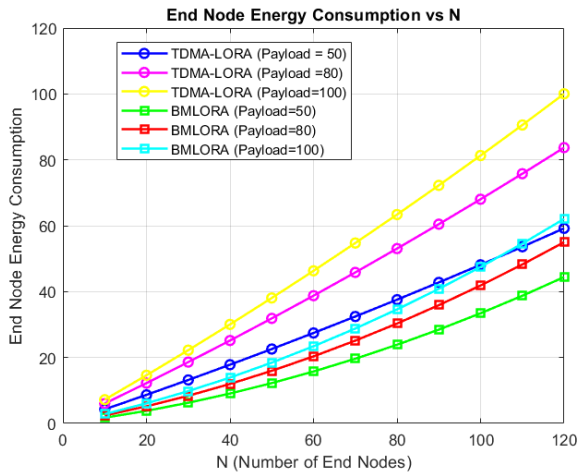

FIGURE 12. End node energy consumption(μJ) vs N (Variation in Payload).

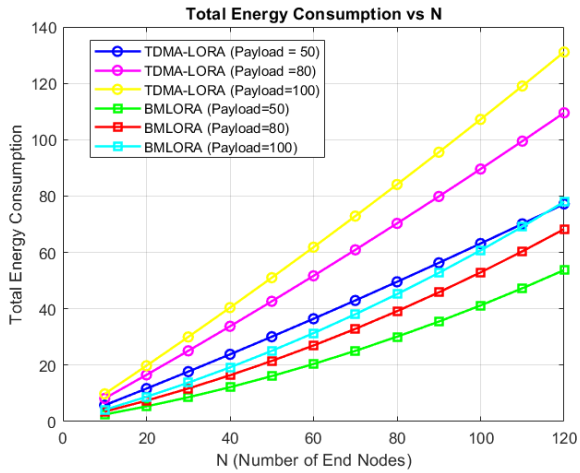

FIGURE 13. Total energy consumption(μJ) vs N (Variation in Payload).

BM-LoRa-based approach (Payload = 100) exhibits energy consumption comparable to the TDMA-based approach (Payload = 50), especially with a 100% increase in throughput.

In the case of varying payload length, AES-256 encryption shows a higher energy consumption compared to AES-128, as expected due to the increased computational overhead of AES-256's additional encryption rounds. However, the difference in energy consumption between AES-128 and AES-256 remains small, particularly when compared to the significant energy savings of the BM-LoRa approach. Even with larger payloads and AES-256 encryption, BM-LoRa continues to outperform the TDMA-based approach regarding energy efficiency.

In all the cases, the number of nodes are varied with different analysis parameters. The results indicate that as node density increases, the bit-mapping-based slot allocation effectively minimizes the energy consumption by reducing contention and energy overhead. The probability variation in Case-1 indicates that the adaptive scheduling mechanism ensures that even under high traffic loads and bursty data traffic, energy consumption remains lower compared to conventional TDMA-based approaches, demonstrating the scalability and robustness of the proposed protocol.

V. CONCLUSION

An energy-efficient security-aware MAC protocol is proposed for efficient data transmission. The performance of the proposed method is compared with the standard LoRa-based TDMA MAC protocol. The analysis shows that the proposed BM-LORA protocol shows significant energy efficiency improvements over TDMA-LORA across various parameters. In terms of energy consumption for varying numbers of nodes, BM-LORA consistently outperforms TDMA-LORA, with energy savings becoming more significant as the number of nodes increases. At 120 nodes, BM-LORA with $p = 0.2$ achieves a reduction of over 70% in total energy consumption as compared to TDMA-LORA. Similarly, when evaluating energy consumption based on payload size, BM-LORA maintains significantly lower energy usage. For a payload size of 100 bytes, BM-LORA reports approximately 60-70% energy savings compared to TDMA-LORA, regardless of the number of nodes in the system. The results clearly indicate that the performance of the proposed method (BM-LoRa) is superior in terms of energy efficiency. Therefore, it can be concluded that BM-LoRa is a robust choice for energy-critical wireless communication applications. In the future, a data encryption-based mathematical model can be developed with enhanced data encryption techniques to strengthen security while maintaining efficient low-power communication in LoRa systems.

REFERENCES

[1] D. Hunt, "LoRa alliance certification," *J. ICT Standardization*, vol. 9, no. 1, pp. 13–20, Apr. 2021, doi: 10.13052/jicts2245-800X.912.

[2] *Platform for IoT*. Accessed: Nov. 13, 2024. [Online]. Available: https://www.semtech.com/vlora

[3] S. Aguilar, D. S. W. La-Torre, A. Platis, R. Vidal, C. Gomez, S. Céspedes, and J. C. Zúñiga, "Packet fragmentation over sigfox: Implementation and performance evaluation of SCHC ACK-on-error," *IEEE Internet Things J.*, vol. 9, no. 13, pp. 11057–11070, Jul. 2022, doi: 10.1109/JIOT.2021.3127170.

[4] *Sigfox 0G Technology by UnaBiz*. Accessed: Nov. 14, 2024. [Online]. Available: https://www.sigfox.com/

[5] *Weightloss Open*. Accessed: Nov. 14, 2024. [Online]. Available: http://www.weightless.org/

[6] M. A. Al-Shareeda, A. Abdullah Alsadhan, H. H. Qasim, and S. Manickam, "Long range technology for Internet of Things: Review, challenges, and future directions," *Bull. Electr. Eng. Informat.*, vol. 12, no. 6, pp. 3758–3767, Dec. 2023.

[7] *Dedicated Machine Connectivity With IoT*. Accessed: Nov. 14, 2024. [Online]. Available: https://www.ingenu.com/

[8] Y.-P. E. Wang, X. Lin, A. Adhikary, A. Grovlen, Y. Sui, Y. Blankenship, J. Bergman, and H. S. Razaghi, "A primer on 3GPP narrowband Internet of Things," *IEEE Commun. Mag.*, vol. 55, no. 3, pp. 117–123, Mar. 2017, doi: 10.1109/MCOM.2017.1600510CM.

[9] L. Vangelista, "Frequency shift chirp modulation: The LoRa modulation," *IEEE Signal Process. Lett.*, vol. 24, no. 12, pp. 1818–1821, Dec. 2017, doi: 10.1109/LSP.2017.2762960.

[10] E. Aras, G. S. Ramachandran, P. Lawrence, and D. Hughes, "Exploring the security vulnerabilities of Lora," in *Proc. 3rd IEEE Int. Conf. Cybern. (CYBCONF)*, Jun. 2017, pp. 1–12, doi: 10.1109/CYBCONF.2017.7985777.

[11] K. Grgic, A. Pejkovic, M. Zrnic, and J. Spisic, "An overview of security aspects of IoT communication technologies for smart agriculture," in *Proc. 16th Int. Conf. Telecommun. (ConTEL)*, Jun. 2021, pp. 146–151, doi: 10.23919/contel52528.2021.9495985.

[12] T.-P. Dang, T.-K. Tran, T.-T. Bui, and H.-T. Huynh, "LoRa gateway based on SoC FPGA platforms," in *Proc. Int. Symp. Electr. Electron. Eng. (ISEE)*, Apr. 2021, pp. 48–52, doi: 10.1109/isee51682.2021.9418711.

[13] M. Eldefrawy, I. Butun, N. Pereira, and M. Gidlund, "Formal security analysis of LoRaWAN," *Comput. Netw.*, vol. 148, pp. 328–339, Jan. 2019, doi: 10.1016/j.comnet.2018.11.017.

[14] F. L. Coman, K. M. Malarski, M. N. Petersen, and S. Ruepp, "Security issues in Internet of Things: Vulnerability analysis of LoRaWAN, sigfox and NB-IoT," in *Proc. Global IoT Summit (GIoTS)*, Jun. 2019, pp. 1–6, doi: 10.1109/giots.2019.8766430.

[15] J. Qadir, J. E. U. Cabus, I. Butun, R. Lagerström, P. Gastaldo, and D. D. Caviglia, "Analysis of LPWAN: Cyber-security vulnerabilities and privacy issues in LoRaWAN, Sigfox, and NB-IoT," in *Low-Power Wide-Area Networks: Opportunities, Challenges, Risks and Threats*, I. Butun and I. F. Akyildiz, Eds., Cham, Switzerland: Springer, 2023, pp. 139–170, doi: 10.1007/978-3-031-32935-7_5. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-031-32935-7_5#citeas

[16] G. Stanco, A. Navarro, F. Frattini, G. Ventre, and A. Botta, "A comprehensive survey on the security of low power wide area networks for the Internet of Things," *ICT Exp.*, vol. 10, no. 3, pp. 519–552, Jun. 2024, doi: 10.1016/j.icte.2024.03.003.

[17] C. Zhang, J. Yue, L. Jiao, J. Shi, and S. Wang, "A novel physical layer encryption algorithm for Lora," *IEEE Commun. Lett.*, vol. 25, no. 8, pp. 2512–2516, Aug. 2021, doi: 10.1109/LCOMM.2021.3078669.

[18] K.-L. Tsai, Y.-L. Huang, F.-Y. Leu, I. You, Y.-L. Huang, and C.-H. Tsai, "AES-128 based secure low power communication for LoRaWAN IoT environments," *IEEE Access*, vol. 6, pp. 45325–45334, 2018, doi: 10.1109/ACCESS.2018.2852563.

[19] A. Iqbal and T. Iqbal, "Low-cost and secure communication system for remote micro-grids using AES cryptography on ESP32 with LoRa module," in *Proc. IEEE Electr. Power Energy Conf. (EPEC)*, Oct. 2018, pp. 1–5, doi: 10.1109/EPEC.2018.8598380.

[20] A. Allamanda, B. W. Hartejo, M. N. Zulfikar, and D. Ogi, "Implementation of AES-256 algorithm for secure data transmission in LoRa-based forest fire monitoring system," in *Proc. IEEE Int. Conf. Cryptography, Informat., Cybersecurity (ICoCICs)*, Aug. 2023, pp. 224–229, doi: 10.1109/ICOCICS58778.2023.10276499.

[21] K.-L. Tsai, F.-Y. Leu, I. You, S.-W. Chang, S.-J. Hu, and H. Park, "Low-power AES data encryption architecture for a LoRaWAN," *IEEE Access*, vol. 7, pp. 146348–146357, 2019, doi: 10.1109/ACCESS.2019.2941972.

[22] K.-L. Tsai, F.-Y. Leu, L.-L. Hung, and C.-Y. Ko, "Secure session key generation method for LoRaWAN servers," *IEEE Access*, vol. 8, pp. 54631–54640, 2020, doi: 10.1109/ACCESS.2020.2978100.

[23] P. Thaenkaew, B. Quoitin, and A. Meddahi, "Leveraging larger AES keys in LoRaWAN: A practical evaluation of energy and time costs," *Sensors*, vol. 23, no. 22, p. 9172, Nov. 2023, doi: 10.3390/s23229172.

[24] P. A. Windya, V. Suryani, and A. A. Wardana, "Sniffing prevention in LoRa network using combination of advanced encryption standard (AES) and message authentication code (MAC)," in *Proc. Int. Conf. Advancement Data Sci., E-Learn. Inf. Syst. (ICADEIS)*, Oct. 2021, pp. 1–5, doi: 10.1109/icadeis52521.2021.9702081.

[25] N. Rakhaine, I. Ahmed, and T. Islam, "A comparative analysis of power consumption and security features in LoRa and LoRaWAN messaging devices," in *Proc. 3rd Int. Conf. Advancement Electr. Electron. Eng. (ICAEEE)*, Apr. 2024, pp. 1–6, doi: 10.1109/icaeee62219.2024.10561782.
[26] S. Abboud and N. Abdoun, "Enhancing LoRaWAN security: An advanced AES-based cryptographic approach," *IEEE Access*, vol. 12, pp. 2589–2606, 2024, doi: 10.1109/ACCESS.2023.3348416.
[27] B. McWeeney, I. Mudritskiy, and R. Verbruggen, "Analysis of payload confidentiality for the IoT/LPWAN technology 'Lora,'" in *Proc. 10th Int. Conf. Inf. Syst. Secur. Privacy*, 2024, pp. 90–102, doi: 10.5220/0012271400003648.
[28] P. S. Munoz, N. Tran, B. Craig, B. Dezfouli, and Y. Liu, "Analyzing the resource utilization of AES encryption on IoT devices," in *Proc. Asia–Pacific Signal Inf. Process. Assoc. Annu. Summit Conf. (APSIPA ASC)*, Honolulu, HI, USA, Nov. 2018, pp. 1200–1207, doi: 10.23919/APSIPA.2018.8659779.
[29] Cypress Semiconductor. *CYW943907AEVAL1F Evaluation Kit*. Accessed: Nov. 13, 2024. [Online]. Available: http://www.cypress.com/documentation/development-kitsboards/cyw943907aeval1f-evaluation-kit
[30] J. C. Liando, A. Gamage, A. W. Tengourtius, and M. Li, "Known and unknown facts of LoRa: Experiences from a large-scale measurement study," *ACM Trans. Sensor Netw.*, vol. 15, no. 2, pp. 1–35, May 2019, doi: 10.1145/3293534.
[31] *Salesforce, SX1301, Semtech.my.salesforce.com*. Accessed: Dec. 12, 2024. [Online]. Available: https://semtech.my.salesforce.com/sfc/p/#E0000000JelG/a/44000000MDnR/Et1KWLCuNDI6MDagfSPAvqqp.Y869Flgs1LleWyfjDY
[32] *Salesforce, SX1276, Semtech.my.salesforce.com*. Accessed: Dec. 12, 2024. [Online]. Available: https://semtech.my.salesforce.com/sfc/p/#E0000000JelG/a/2R0000001Rbr/6EfVZUorrpoKFfvaF_Fkpgp5kzjiNyiAbqcpqh9qSjE

K. TUSHAR VIKASH (Member, IEEE) was born in Coimbatore, Tamil Nadu, India, in November 2004. He is currently pursuing the B.Tech. degree in computer and communication engineering with the Department of Information and Communication Technology, Manipal Institute of Technology, Manipal, Karnataka, India. In 2024, he completed an internship as the IoT Architect at ConnectME Manufacturing LLC, Dubai, where he involved on designing the IoT-based solutions for smart manufacturing and optimization applications. His research interests include LoRa, energy-efficient network protocols, the IoT MAC layer protocols, machine learning applications in embedded systems and communication, and enhancing the performance and efficiency of low-power communication systems, with an emphasis on integrating machine learning for optimization.

GAURAV SUMAN received the B.E. degree in electronics and communication engineering from Annamalai University, in 2013, and the M.Tech. degree in data science and engineering from the Birla Institute of Technology and Science (BITS), Pilani, in 2023. He is currently pursuing the Ph.D. degree with the Department of Information and Communication Technology, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal. He is with Visteon Technical and Services Center Pvt., Ltd., and utilizes engineering skills in automotive development. With over ten years of experience in the automotive industry, he has previously worked for companies, such as HCL Technologies, KPIT Technologies, and Tata Consultancy Services before joining Visteon. His research interests include wireless communication, the Internet of Things, and image segmentation.

MANOJ TOLANI (Senior Member, IEEE) received the B.Tech. degree from the IIMT Engineering College, Meerut, India, in 2010, the M.Tech. degree from the Department of Electronics and Communication Engineering, Madan Mohan Malaviya University of Technology, Gorakhpur, India, in 2012, and the Ph.D. degree from the Department of Electronics and Communication Engineering, Indian Institute of Information Technology Allahabad, India. He was an Assistant Professor with the PSIT College of Engineering, Kanpur, from 2012 to 2015. He was a Teaching Cum Research Associate (TRA) with the Department of ECE, IIIT-Allahabad, in 2020. He was with the Atria Institute of Technology, Bangalore, as an Assistant Professor (Research), from 2021 to 2023. He is currently with Manipal Institute of Technology, MAHE, Manipal. He has published more than 20 journals and conferences. His research interests include wireless sensor networks, the Internet of Things, and software computing.

AKSHAY TEJA TUMUNURI (Member, IEEE) is currently pursuing the B.Tech. degree in computer and communications engineering with the Department of Information and Communication Technology, Manipal Institute of Technology, Manipal, Karnataka, India. His research interests include networking, the Internet of Things, cloud computing, and machine learning. In 2024, he undertook an internship at Navmarg Research and Innovation Pvt., Ltd., as a Full Stack Web Developer, where he was responsible for developing the company's website and contributing to the creation of various products.

PANKAJ KUMAR (Member, IEEE) received the B.Tech. degree (Hons.) in electronics and communication engineering and the M.Tech. degree (Hons.) in digital communication from Dr. A. P. J. Abdul Kalam Technical University, formerly Uttar Pradesh Technical University, Lucknow, India, and the Ph.D. degree from Indian Institute of Technology (IIT) Ropar, India. He was a Postdoctoral Fellow with Indian Institute of Science (IISc), Bengaluru, India. He is currently an Assistant Professor with the Department of Information and Communication Technology, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, India. His research interests include drone assisted cooperative networks, network coding, network coded cooperation, channel correlation, D2D communication, energy harvesting, and intelligent reflecting surface, and NOMA.

Open Access funding provided by 'Manipal Academy of Higher Education' within the CRUI CARE Agreement